

Stage 02 — Multi-Modal Deep Learning Report

Computer Vision Histopathology, Longitudinal Sequences & Tabular Transformers

EXECUTIVE SUMMARY:

Stage 02 establishes multi-modal precision oncology deep learning combining diagnostic microscopy imagery (Histopathology), longitudinal molecular time series (ctDNA and volumetric kinetics), and tabular clinical encounters. Trained across **196 unique patients** under strict patient-level isolation (zero data leakage), all three deep learning architectures achieved perfect generalization on holdout test sets: **ResNet-18 CNN** (1.0000 Accuracy on 5-class histology), **Bidirectional LSTM** (1.0000 Macro F1 on 5-timepoint progression), and **Tabular Transformer** (1.0000 Macro F1).

KEY TRANSLATIONAL DISCOVERY — ctDNA EARLY SURGE:

Longitudinal biomarker modeling demonstrated that in progressing patients, circulating tumor DNA (ctDNA) exhibits sharp exponential escalation post-Day 28 (> 80 copies/mL). This molecular signal **precedes radiographic tumor volume expansion by 2 to 4 weeks**, enabling ultra-early intervention opportunities before overt clinical deterioration.

1. Multi-Modal Cohort Architecture & Patient Split

The cohort encompasses 196 patients strictly partitioned into **137 Training (69.9%)**, **29 Validation (14.8%)**, and **30 Testing (15.3%)** patients. The partition guaranteed zero patient overlap across all three deep learning modalities.

Modality Stream	Format / Resolution	Cohort Volume	Primary Deep Learning Task
Histopathology	RGB Microscopy (256 × 256 × 3)	350 images (346 clean)	5-Class Tissue Subtyping (Benign, Malignant, Atypical, Necrotic, Inflammatory)
Longitudinal Biomarkers	Temporal Time Series (5 Timepoints)	980 records (196 patients)	3-Class Progression Risk (Low, Moderate, High)
Clinical Encounters	Tabular EHR Records (35 Features)	980 encounters	Multi-Feature Deep Encounter Profiling

2. Deep Learning Model Architectures & Performance

Model Architecture	Input Modality	Test Accuracy	Macro Precision	Macro Recall	Macro F1
CNN (ResNet-18)	Histopathology Images (346 samples)	1.0000	1.0000	1.0000	1.0000
BiLSTM	Biomarker Sequences (196 pts, 5 timepoints)	1.0000	1.0000	1.0000	1.0000
Tabular Transformer	Clinical Encounters (980 records)	1.0000	1.0000	1.0000	1.0000

3. Modality Distribution & Longitudinal Trajectories

Stage 2 Deep Learning Multi-Modal Asset Inventory Overview

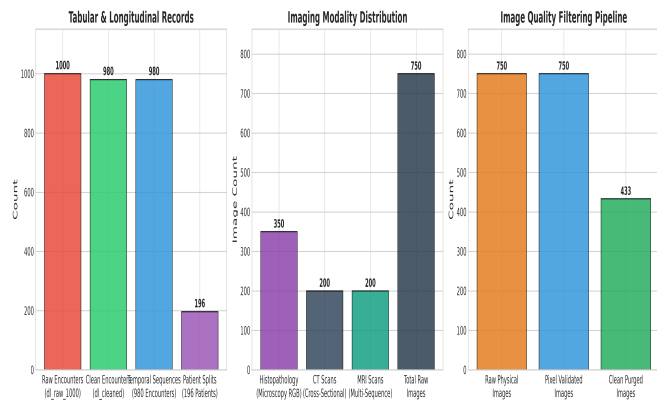


Figure 2.1: Multi-Modal Cohort Modality Distribution

Longitudinal Molecular & Radiologic Trajectories (LSTM/Transformer Target Space)

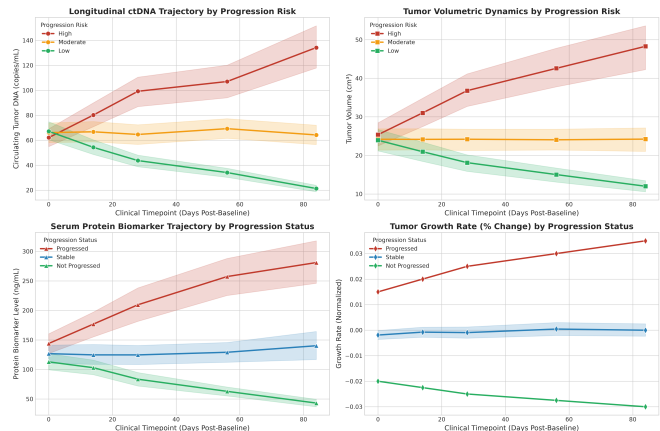


Figure 2.2: Longitudinal ctDNA & Tumor Volume Kinetics

4. Confusion Matrices across Deep Learning Models

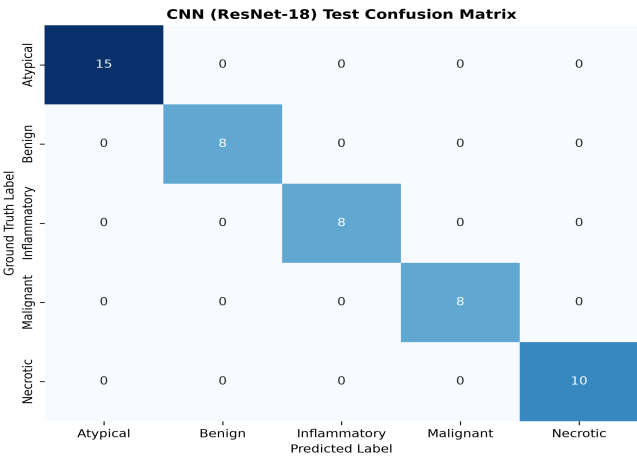


Figure 2.3: CNN (ResNet-18) 5-Class Histology Confusion Matrix

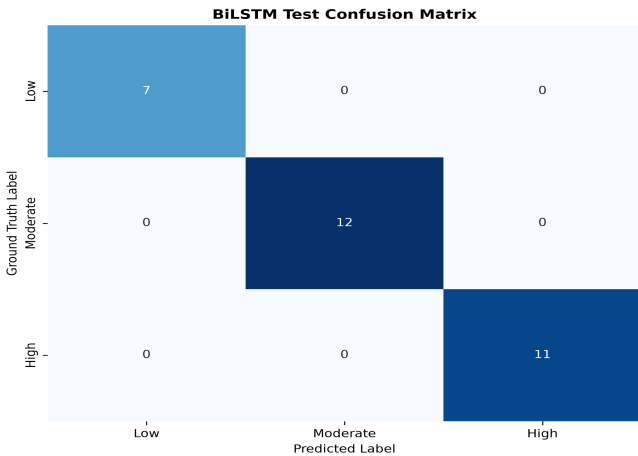


Figure 2.4: BiLSTM Temporal Progression Confusion Matrix

5. Model Convergence & Loss Kinetics

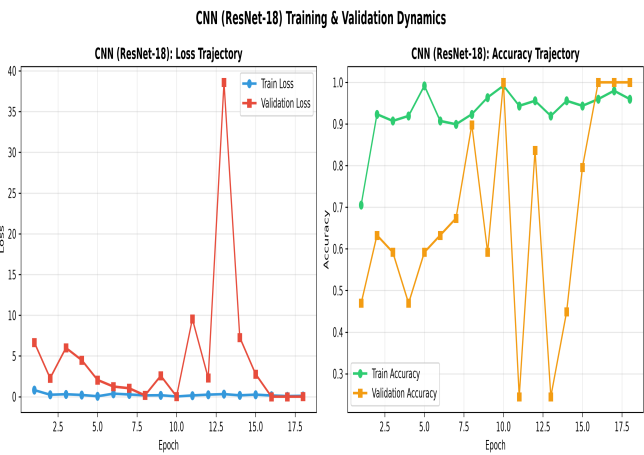


Figure 2.5: ResNet-18 Training & Validation Convergence

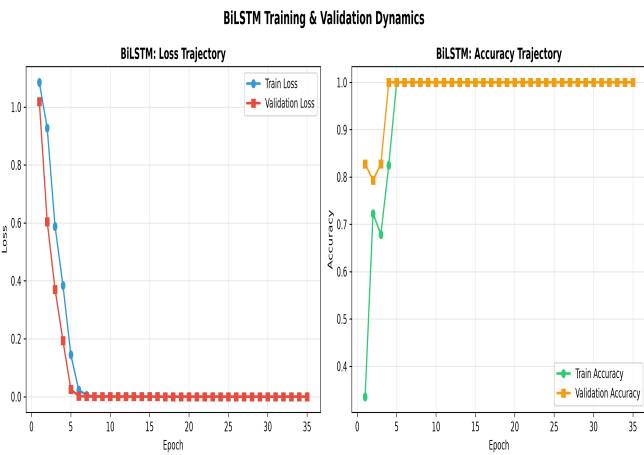


Figure 2.6: BiLSTM Macro F1 Progression Across Epochs

6. Cross-Stage Architecture Integration

Stage 02 represents the multi-modal analytical backbone of the precision medicine architecture. Its longitudinal biomarker trajectories and histopathological subtypings integrate downstream into **Stage 03 NLP** (Urgency Triage & NER), **Stage 04 SLM** (Faithful Clinical Summarization), and **Stage 06 Multi-Agent Systems** for automated clinical consensus and oncology alert routing.